WORKED SOLUTION

STEP II 2024, Question 8

Pure •

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 2.6, *The Arithmetic–Geometric Mean*: https://beyondstep.keithmcnulty.org/ch02/agm.html

The key idea

The two sequences squeeze together: the arithmetic means come down, the geometric means go up, and the gap at least halves each time, so both settle on a common limit M, the *arithmetic–geometric mean* of x_0 and y_0. In part (ii) the substitution turns the integral for the pair (p, q) into the integral for the pair of their means, so the integral never changes as we run the iteration. Letting $n \to \infty$ collapses it to an integral we can do.

Solution

Part (i)

We first show by induction that $0 < y_n < x_n$ for every n. It holds for $n = 0$. Suppose $0 < y_n < x_n$. Then $\sqrt{x_n} \neq \sqrt{y_n}$, so

$$0 < \left(\sqrt{x_n} - \sqrt{y_n}\right)^2 = x_n + y_n - 2\sqrt{x_n y_n},$$

which rearranges to $\frac{1}{2}(x_n + y_n) > \sqrt{x_n y_n}$, that is, $y_{n+1} < x_{n+1}$. Also $y_{n+1} = \sqrt{x_n y_n} > 0$. This completes the induction, so $y_{n+1} < x_{n+1}$ for all $n \geq 0$.

Using $y_n < x_n$:

$$x_{n+1} = \tfrac{1}{2}(x_n + y_n) < \tfrac{1}{2}(x_n + x_n) = x_n,$$
$$y_{n+1} = \sqrt{x_n y_n} > \sqrt{y_n y_n} = y_n,$$

where the second line uses $x_n > y_n > 0$ and the fact that the square root is increasing.

So (y_n) is increasing. It is also bounded above: $y_n < x_n < x_{n-1} < \cdots < x_0$, so $y_n < x_0$ for all n. By the theorem, $y_n \to M$ for some $M \leq x_0$. (Since the sequence increases from y_0, also $M \geq y_0 > 0$.)

For the gap, the left-hand inequality $x_{n+1} - y_{n+1} > 0$ is what we proved first. For the right-hand one, since $y_{n+1} > y_n$,

$$x_{n+1} - y_{n+1} < x_{n+1} - y_n = \tfrac{1}{2}(x_n + y_n) - y_n = \tfrac{1}{2}(x_n - y_n).$$

Applying this repeatedly,

$$0 < x_n - y_n < \frac{x_0 - y_0}{2^n} \to 0,$$

so $x_n - y_n \to 0$ by squeezing. Finally $x_n = y_n + (x_n - y_n)$ is the sum of a sequence tending to M and one tending to 0, so $x_n \to M$ as well.

Part (ii)

Write $a = \tfrac{1}{2}(p+q)$, $g = \sqrt{pq}$, and call the t-integral J.

The substitution. With $t = \tfrac{1}{2}\left(x - \frac{pq}{x}\right)$ for $x > 0$,

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{2}\left(1 + \frac{pq}{x^2}\right) = \frac{x^2 + pq}{2x^2} > 0,$$

so t increases strictly, from $-\infty$ as $x \to 0^+$ to $+\infty$ as $x \to \infty$: the substitution maps $(0, \infty)$ one-to-one onto the whole real line. Now we rewrite the two factors under the root.

$$pq + t^2 = \tfrac{1}{4}\left(x^2 + 2pq + \frac{p^2q^2}{x^2}\right) = \frac{1}{4}\left(x + \frac{pq}{x}\right)^2 = \left(\frac{x^2 + pq}{2x}\right)^2,$$

$$\begin{aligned}\tfrac{1}{4}(p+q)^2 + t^2 &= \tfrac{1}{4}\left(p^2 + 2pq + q^2 + x^2 - 2pq + \frac{p^2q^2}{x^2}\right) \\ &= \frac{x^4 + (p^2+q^2)x^2 + p^2q^2}{4x^2} = \frac{(x^2+p^2)(x^2+q^2)}{4x^2}.\end{aligned}$$

Since $x > 0$ and $x^2 + pq > 0$, the square root of the product is

$$\frac{x^2 + pq}{2x} \cdot \frac{\sqrt{(p^2+x^2)(q^2+x^2)}}{2x}.$$

Therefore

$$\begin{aligned}J &= \int_0^\infty \frac{x^2 + pq}{2x^2} \cdot \frac{4x^2}{(x^2+pq)\sqrt{(p^2+x^2)(q^2+x^2)}}\,\mathrm{d}x \\ &= 2\int_0^\infty \frac{\mathrm{d}x}{\sqrt{(p^2+x^2)(q^2+x^2)}} = 2\,\mathrm{I}(p,q).\end{aligned}$$

The same integral read directly. Since $\tfrac{1}{4}(p+q)^2 = a^2$ and $pq = g^2$, the integrand of J is $1/\sqrt{(a^2+t^2)(g^2+t^2)}$, which is even in t. So

$$J = 2\int_0^\infty \frac{\mathrm{d}t}{\sqrt{(a^2+t^2)(g^2+t^2)}} = 2\,\mathrm{I}(a,g),$$

which is legitimate because a, g are positive with $g < a$ (by the inequality of part (i), as $p \neq q$). Comparing the two expressions for J gives

$$\mathrm{I}(p,q) = \mathrm{I}\big(\mathrm{a}(p,q), \mathrm{g}(p,q)\big),$$

as required.

Evaluating $\mathrm{I}(x_0, y_0)$. Applying this with $(p,q) = (x_n, y_n)$, which is allowed since $0 < y_n < x_n$, gives $\mathrm{I}(x_n, y_n) = \mathrm{I}(x_{n+1}, y_{n+1})$, so

$$\mathrm{I}(x_0, y_0) = \mathrm{I}(x_n, y_n) \quad \text{for every } n.$$

Ask yourself We want to let $n \to \infty$ and say the integral tends to $\mathrm{I}(M, M)$. How can we justify moving the limit inside an integral over an infinite range?

We avoid any delicate limit argument by squeezing. For $0 < y_n < x_n$ and every x,

$$(y_n^2 + x^2)^2 \leq (x_n^2 + x^2)(y_n^2 + x^2) \leq (x_n^2 + x^2)^2,$$

so, taking square roots and reciprocals (which reverses the inequalities) and integrating,

$$\int_0^\infty \frac{\mathrm{d}x}{x_n^2 + x^2} \leq \mathrm{I}(x_n, y_n) \leq \int_0^\infty \frac{\mathrm{d}x}{y_n^2 + x^2}.$$

For $c > 0$, $\displaystyle\int_0^\infty \frac{\mathrm{d}x}{c^2 + x^2} = \Big[\frac{1}{c} \arctan \frac{x}{c}\Big]_0^\infty = \frac{\pi}{2c}$, so

$$\frac{\pi}{2x_n} \leq \mathrm{I}(x_0, y_0) \leq \frac{\pi}{2y_n} \quad \text{for every } n.$$

The left-hand side is constant in n, and both bounds tend to $\pi/(2M)$ because $x_n \to M$ and $y_n \to M$ with $M > 0$. Hence

$$\mathrm{I}(x_0, y_0) = \frac{\pi}{2M}.$$

Checks

- Take $x_0 = 2$, $y_0 = 1$. The iteration gives $(1.5, 1.41421)$, then $(1.45711, 1.45648)$, and then both agree to five figures at $M \approx 1.45679$. So $\pi/(2M) \approx 1.07826$, and evaluating $\mathrm{I}(2,1)$ numerically gives 1.07826 too.
- If $p = q$ the identity is trivially true, since then $\mathrm{a}(p,p) = \mathrm{g}(p,p) = p$. The answer $\pi/(2M)$ is consistent with this, because $\mathrm{I}(M,M) = \pi/(2M)$.
- The convergence is extremely fast: after the first few steps the gap roughly squares each time, much better than the halving we proved.

Common slips

- Using $y_{n+1} < x_{n+1}$ in the induction without first establishing $y_n < x_n$ (and $y_n > 0$, so that the square roots make sense).
- Forgetting that the t-integral runs over the whole real line, and so equals *twice* $\mathrm{I}(a, g)$; the two factors of 2 must cancel.
- Writing $\mathrm{I}(x_n, y_n) \to \mathrm{I}(M, M)$ without any justification. The squeeze above gives an honest reason.

Beyond STEP

The common limit M is Gauss's arithmetic–geometric mean, and the integral I is a *complete elliptic integral*: Gauss found this identity in 1799, and it gives an astonishingly fast way to compute elliptic integrals, and even π itself (the Gauss–Legendre algorithm). The monotone convergence theorem quoted at the start, and the care needed when swapping limits and integrals, are central themes of a first course in analysis; Abbott's *Understanding Analysis* is an excellent place to meet them properly.

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